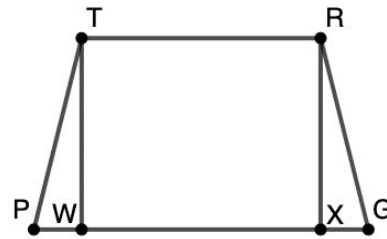


Geometry
Unit 8
Lesson 5 Homework

Name _____
Date _____
Period _____

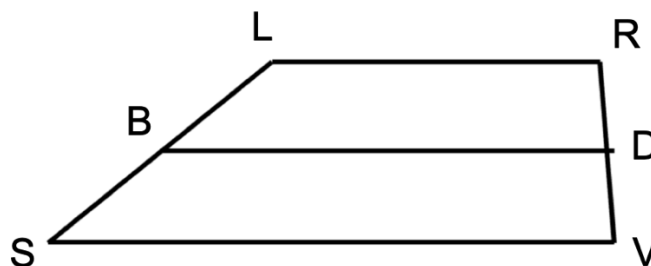
Given: $TRGP$ is an isosceles trapezoid with altitudes $\overline{TW}$ and $\overline{RX}$.

Prove: $\overline{PW} \cong \overline{XG}$



Statement	Reason
$TRGP$ is an isosceles trapezoid	Given
1.	Altitudes of an isosceles trapezoid are congruent.
$\overline{TW} \perp \overline{PG}$ and $\overline{RX} \perp \overline{PG}$	2.
$\angle TWP$ and $\angle RXG$ are right angles.	Perpendicular lines form right angles
3.	All right angles are congruent
4.	5.
6.	AAS Congruence Theorem
$\overline{PW} \cong \overline{XG}$	Corresponding parts of congruent triangles are congruent

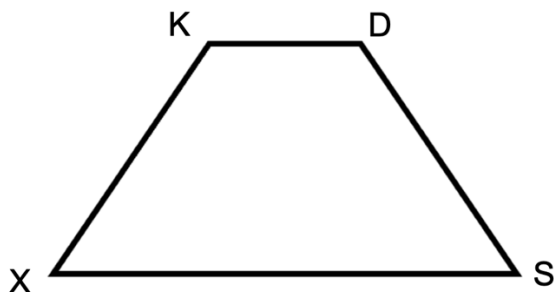
Trapezoid $LRVS$ is shown with the midsegment $\overline{BD}$ where $LB = 5 + 3x$, $SB = 5x - 11$, $RD = 9y + 16$, $VD = 51 + 4y$.



7. What is the value of x and y ?

8. Find the length of $\overline{LB}$ and $\overline{RV}$.

$KDSX$ is an isosceles trapezoid, as shown, where $m\angle K = (35z - 20)^\circ$, $m\angle X = (15z + 8)^\circ$, $m\angle S = (10t + 25)^\circ$.



9. What is the value of z and t ? Round your answer to the nearest hundredth.

10. What is $m\angle S$ and $m\angle D$? Round your answer to the nearest tenth.